CTEC601 2026 S1

Assignment 2 – A Synthetic Reality

Name of the Scene:

Portal Chamber 00

Team Members

Name	Contribution description	Contribution weight	Signature
Kyle Elmo Malimban	Asset Modelling, Custom Texturing, Material Application, VFX/SFX Implementation	50%	
Al Tristan Legaspi	Environment/Map Modelling, Unity Interaction Systems, Scene Layout Refinement, Hierarchy Organisation	50%	
…			

Description

Roughly describe the project:

- ***Why did you decide for the location/scene/etc.?***
  We decided to recreate the Portal 1 starting room because Portal is a very unique and iconic game with a strong visual identity. Even though the map design looks simple at first, it is still very impressive because of how clean, organised, and purposeful the environment feels. The room uses simple shapes, clear wall panels, glass sections, doors, buttons, and test chamber objects, but these elements work together to create a memorable sci-fi testing environment. We thought this would be a good choice for a synthetic reality scene because it allowed us to focus on recreating the atmosphere of Portal through modelling, texturing, lighting, and interactions, while still keeping the scene achievable within the assignment timeframe. The Portal starting room also gave us opportunities to include interactive elements such as sliding doors, buttons, cube physics, digital displays, and particle effects, which helped make the environment feel more alive and closer to the original game.

- ***What core assets did you need for this project?***
  The core assets needed for this project included the main Portal-style chamber environment, such as the wall panels, floor tiles, ceiling tiles, glass room, metal frames, doorways, and orange light strips. We also needed key Portal-inspired objects such as the companion cube, cube dropper, pressure button, small dropper button, circular portal doors, digital display screen, Aperture logo/signage, CCTV camera, table, radio, mug, toilet, and sleeping pod. In addition to the physical models, we also needed custom textures and materials for the concrete walls, tiled floors, roof panels, glass, metal trims, glowing light strips, and darker inset wall details. For the interactive part of the scene, we needed scripts

and Unity components for the sliding doors, cube physics, cube pickup, pressure button, cube dropper button, digital display flicker, particle sparks, and sound effects.

- ***How did you record/create/process those core assets?***
  We used Portal 1 gameplay/reference images to study the layout, colours, shapes, and overall design of the starting room. The environment was first blocked out in Unity using primitive shapes so we could get the main room layout, spacing, and proportions correct. After that, custom assets and textures were created in Blender, including wall/floor textures, door frames, buttons, the companion cube, digital display frame, and other room props. The textures were processed into usable Unity materials, including albedo, normal, height, and roughness-style maps where needed. After importing the models and textures into Unity, we adjusted their scale, positioning, material tiling, lighting, and colliders so they worked correctly in the scene. We also processed the interactive elements inside Unity by adding physics components, triggers, particle systems, audio sources, and scripts to make the scene feel more dynamic and closer to the original Portal environment.

  Sound Creation and Audio Sources
  Most of the sound effects used in the scene were created by recording our own audio using a phone voice recorder. These included sounds such as button press effects and Portal sliding door opening and closing sounds. The recordings were then processed and edited to better fit the Portal-style environment, such as trimming the clips, adjusting volume, and making them feel more mechanical or sci-fi before importing them into Unity.

  The user movement and footstep sounds were taken from the audio resources provided through the course lecture materials, as these were part of the supplied Unity controller setup.

  For sounds that were more difficult to recreate accurately ourselves, such as electrical spark/crackling effects for the portal door sparks and the blue elevator particle energy hum, we used online sound resources instead. This was because these types of electrical and sci-fi energy sounds were harder to safely and realistically recreate through DIY recording. These external sounds were only used where necessary and have been cited below.

  External Sound Sources Used

  - Electrical spark/crackle sound effect: *Electric Sparks* by **freesound_community** from **Pixabay**.
    Source link: https://pixabay.com/sound-effects/search/sparks/
    Reason for use: This sound was difficult to recreate safely and realistically through DIY recording, so an online sound resource was used and cited.

  - Blue elevator particle/energy hum sound effect: electric hum sound effect from **Pixabay**.
    Source link: https://pixabay.com/sound-effects/search/electric-hum/
    Reason for use: This sound was used to support the blue elevator scan-line particle effect. It was difficult to recreate realistically through DIY recording because it required a consistent sci-fi electrical hum, so an online sound resource was used and cited.
- ***Did you break down the project into steps/blocks? If so, which?***
  Yes, we broke the project down into separate stages to make the scene easier to build and

manage. First, we focused on gathering Portal 1 reference images and blocking out the main room layout in Unity using primitive objects. This helped us plan the proportions of the spawn room, first room, glass section, and additional test chamber areas before adding detailed assets.

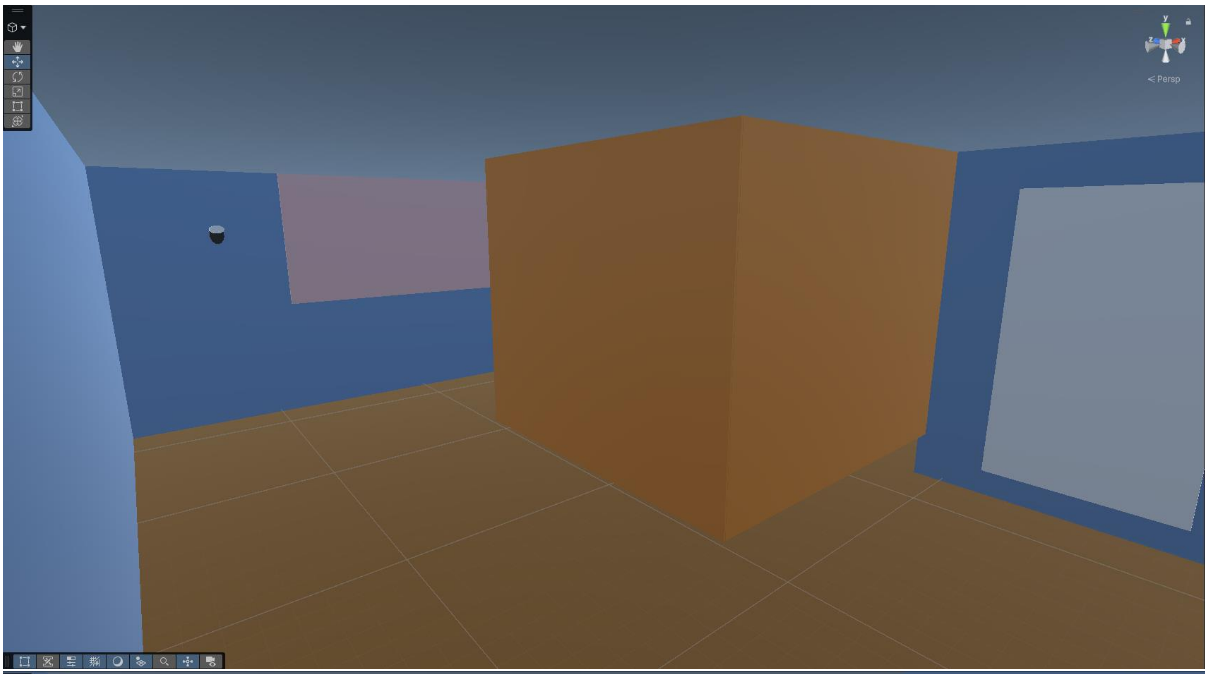
After the blockout was completed, we worked on creating the main environment assets, including wall panels, floor tiles, ceiling tiles, glass frames, door frames, and room props. Blender was used to create several custom assets and textures, while Unity was used to arrange the objects, apply materials, and test how everything fit together in the scene.

The next stage focused on interactions and dynamic elements. This included the Portal-style sliding doors, trigger areas, companion cube physics, cube pickup system, pressure button, cube dropper button, digital display flicker, particle sparks, and sound effects. These features were tested in Unity to make sure they worked correctly and made the scene feel more interactive.

Finally, we focused on polishing the scene by adjusting lighting, material tiling, object scale, colliders, hierarchy organisation, and audio/visual effects. This helped make the final synthetic reality feel more coherent and closer to the Portal reference environment.

Sketches/Screenshots

Screenshots from prototype version: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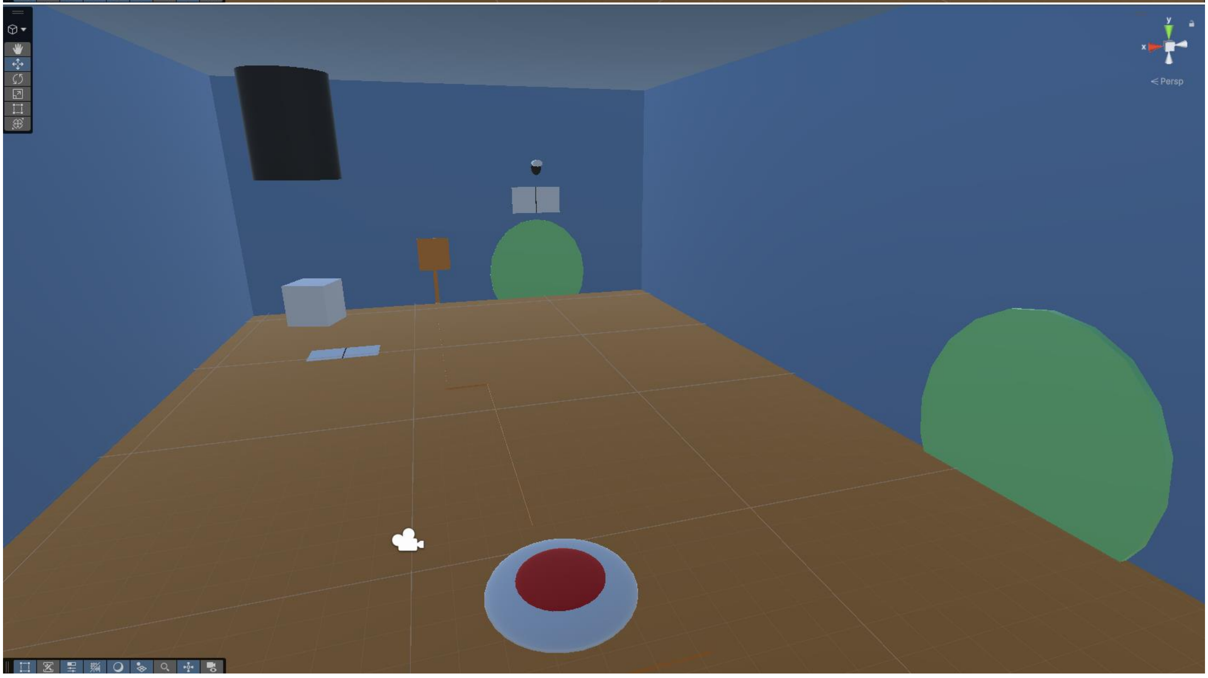
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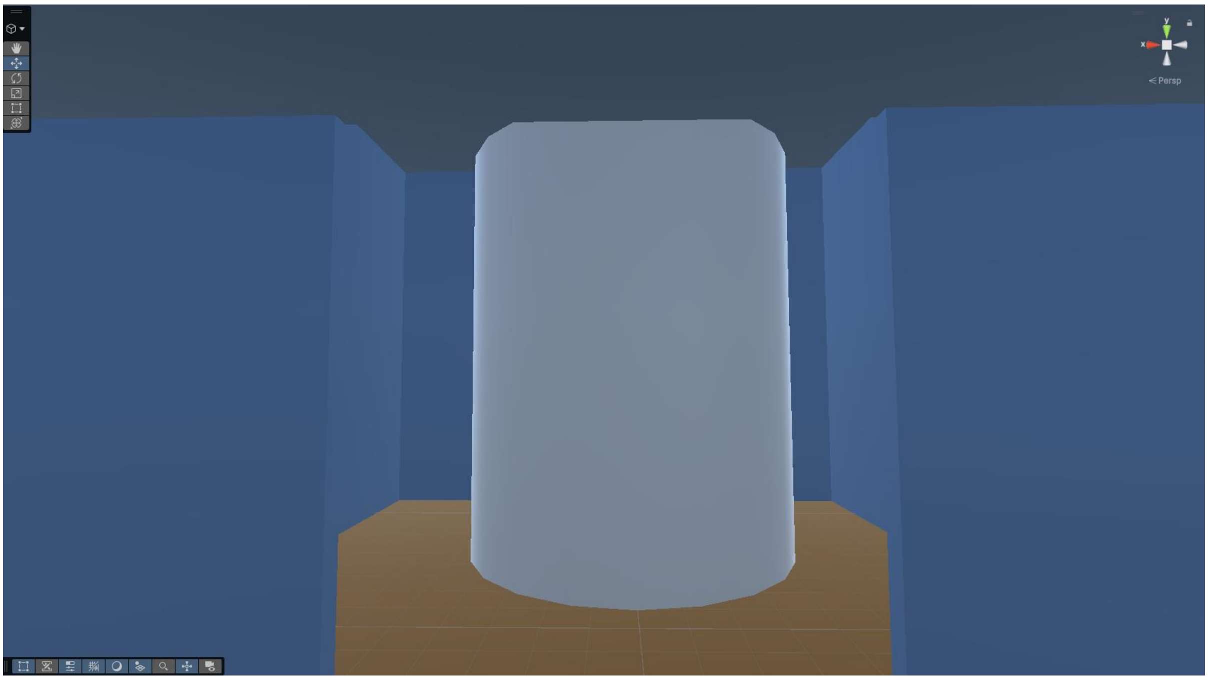
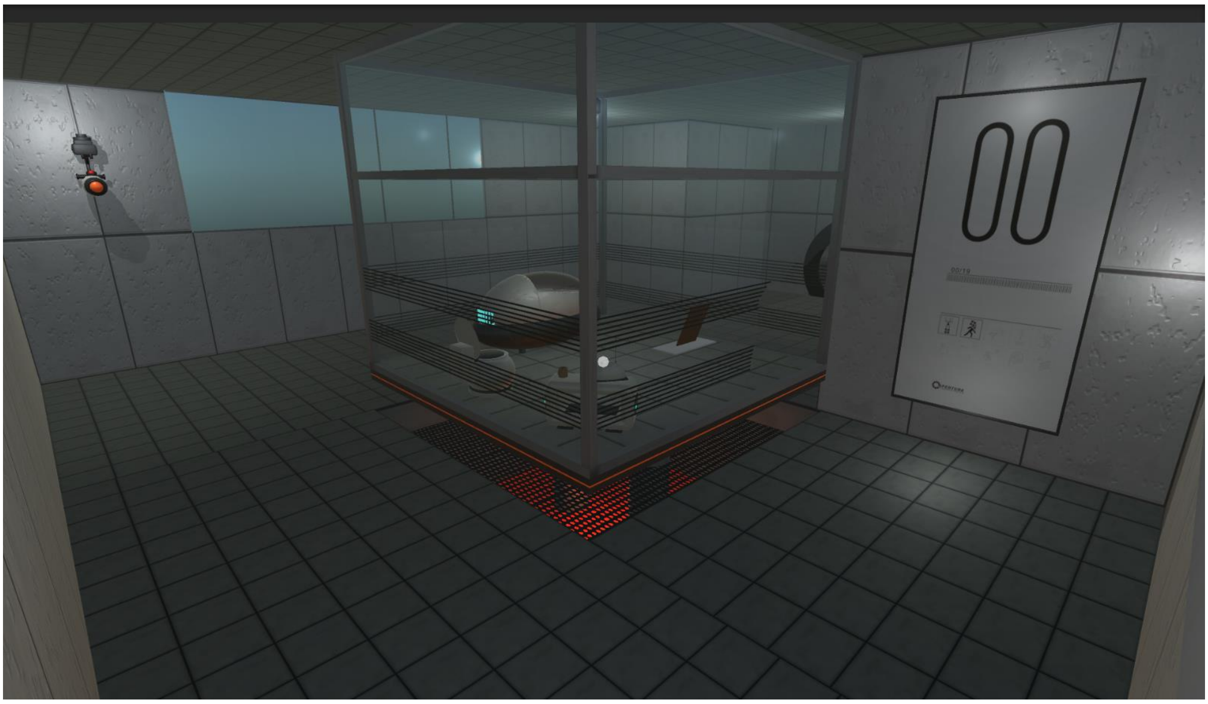
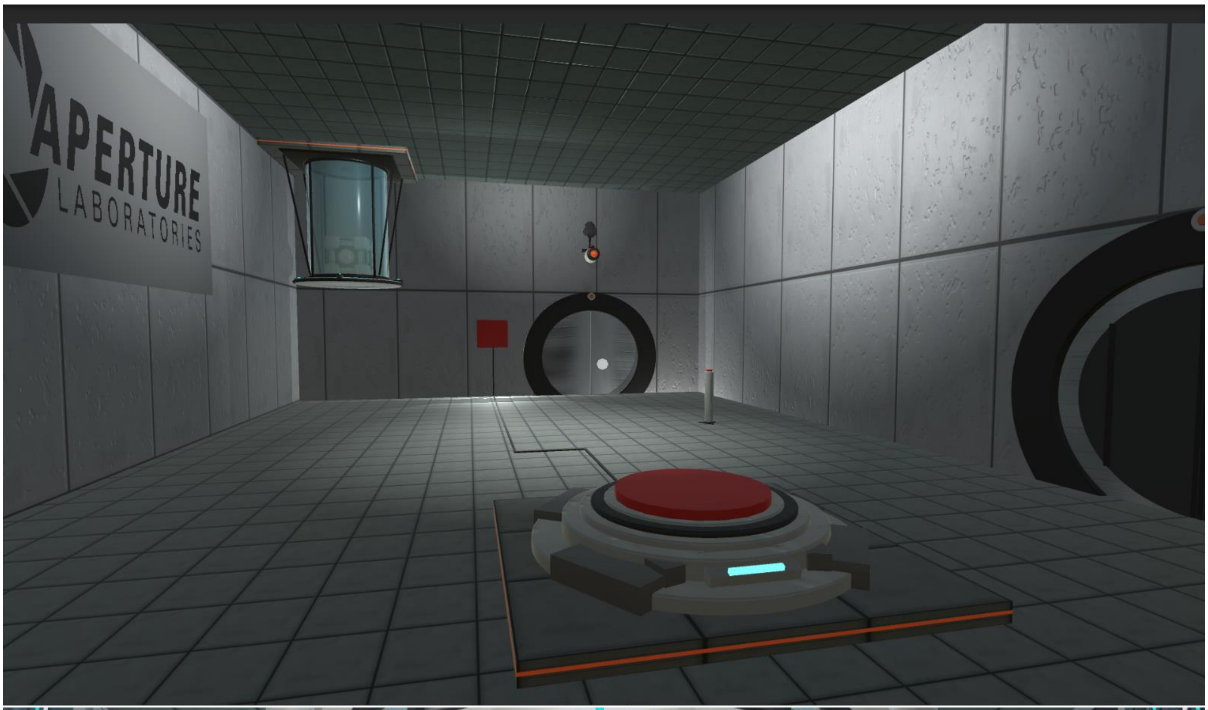
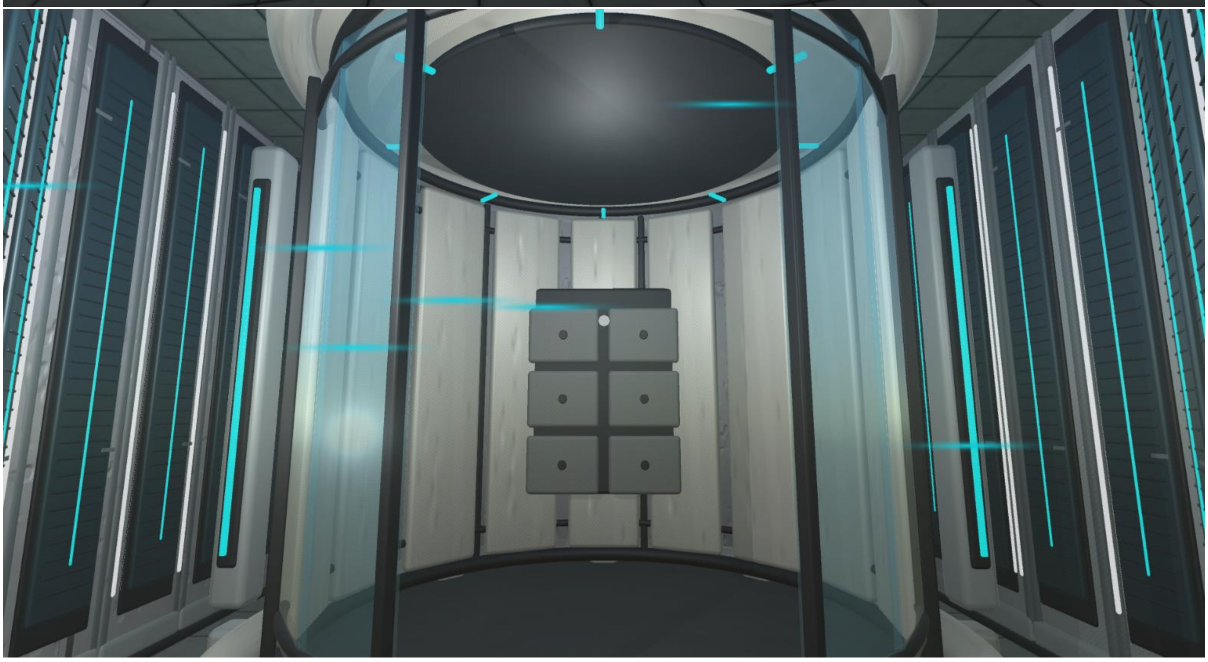
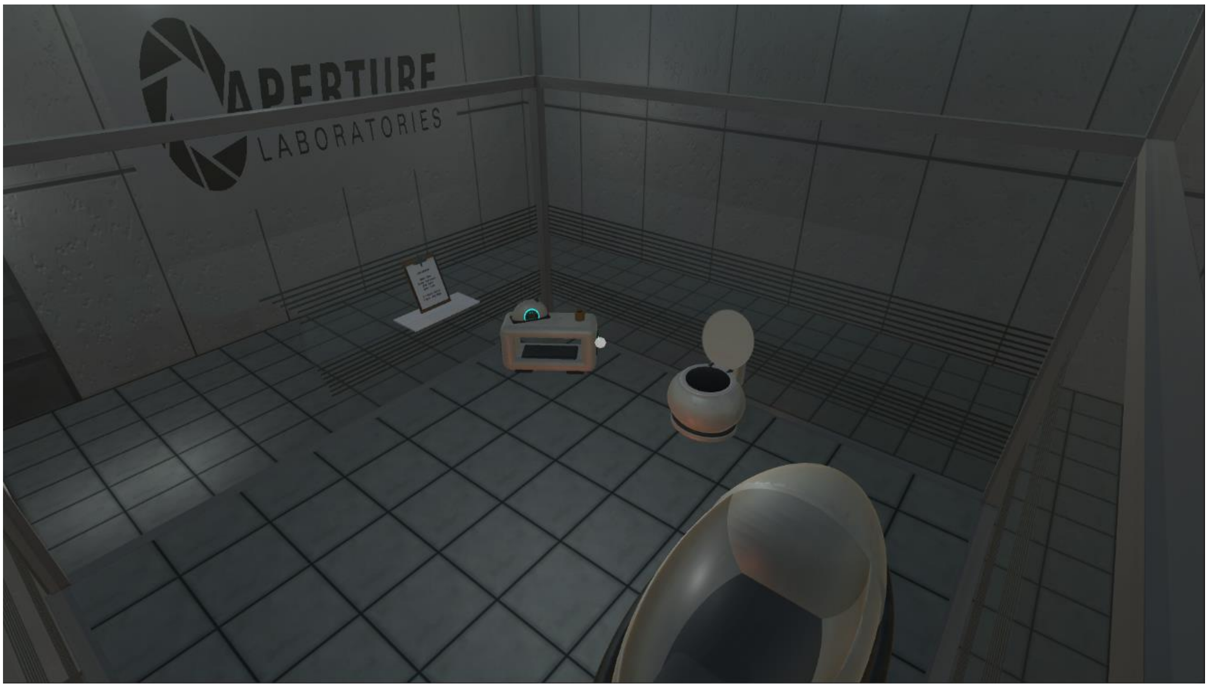


Screenshots from final version:




APERTURE
LABORATORIES





Changelog/Diary

21/04/2026 - Both

- Assessment 2 introduced in class
- Found group members
- Brainstormed possible ideas for the Synthetic Reality

28/04/2026 - Both

- Finalized our idea for the assignment – both agreed upon doing the Portal 1 starting room
- Opened the assignment template, began brainstorming possible ways to get reference photos

7/05/2026 - Both

- Started on the assignment, took reference photos in-game and discussed ideas for the assignment.
- Both received a copy of the assignment template for our individual contributions, similar to how it was done in Assignment 1.

14/05/2026 - Tristan

- Created the prototype model of the synthetic reality using primitive objects and colours, made the second room, third room, spawn room and first room.

19/05/2026 - Both

- Installed Blender and familiarized ourselves with its UI and texturing features

26/05/2026 - Kyle

- Created and tested custom wall and floor textures based on the Portal 1 starting room reference images.

- Used Blender UV mapping to prepare the wall/floor textures so they could be applied correctly in Unity.
- Designed the wall texture with light concrete panels, dark seam lines, and subtle stain details to match the Portal chamber style.
- Imported the textures into Unity, created materials, and applied them to the chamber walls and floor.

27/05/2026 - Tristan

- Updated the naming conventions of the objects as well as cleaning up the object hierarchy
- Duplicated the original wall texture for each wall object and adjusted the material tiling in Unity to make the panels look more proportional, reduce stretching on different wall sizes and proper alignment with neighbouring walls/surfaces

28/05/2026 - Kyle

- Created and tested the glass room area using transparent materials, metal frames, and dark horizontal glass lines to better match the Portal reference room.
- Adjusted the overall room lighting, skybox/environment lighting, and point lights to make the chamber less dark and closer to the example video.

29/05/2026 - Kyle

- Created the raised square room base, orange glowing light strip, and bottom grate area underneath the glass room.
- Tested different floor layouts and added a square floor gap to allow extra detail under the room, similar to the Portal reference scene.

30/05/2026 - Kyle

- Used Blender to create and export custom assets for the scene, including the broken glass panel, button base, companion cube, and digital display frame.
- Imported the Blender models into Unity and adjusted their scale, position, materials, and layout to fit inside the Portal starting room.

31/05/2026 - Kyle

- Created additional wall and ceiling textures in Blender, including concrete panel textures, roof tiles, and dark inset wall panel details.
- Applied the new textures in Unity and adjusted the material tiling so the panels looked more proportional across different wall and ceiling sizes.

1/06/2026 - Kyle

- Created and tested the circular door frame area using Blender and Unity, including a wall section with a door opening and matching Portal-style frame.

3/06/2026 - Tristan

- Tested the player controller, camera setup, floor colliders, and object placement to make sure the player could walk around the scene correctly.
- Cleaned up hierarchy tab – following set naming conventions and putting all objects under the appropriate parent objects

3/06/2026 - Tristan

- Created and tested the Portal-style split door interaction, where the two door panels slide open when the player activates a trigger.
- Added a second trigger on the opposite side of the door so it closes behind the player after walking through.

3/06/2026 - Tristan

- Added the main interactive systems for the scene, including the pressure button, companion cube physics, cube pickup, and cube dropper button.
- Connected the companion cube and pressure button to the second portal door so the door opens when the cube is placed on the button and closes when it is removed.
- Created the digital display interaction, where the screen starts black, waits briefly, then flickers on with the Portal-style display image.
- Added particle sparks and sound effects for the portal doors, buttons, and electrical malfunction effects to make the scene feel more dynamic and interactive.

4/06/2026 - Kyle

- Fully created the elevator room section of the Portal scene, including the main elevator structure and surrounding layout so it matched the intended Portal-inspired design.
- Added the surrounding elevator wall panels and extra scene details to make the area feel more complete and visually accurate.
- Added additional visual effects for the elevator area, including cyan scan-line particle effects, to better match the Portal reference scene and improve the sci-fi atmosphere.
- Adjusted the particle material, colour, movement, and placement so the elevator section feels more dynamic and visually connected to the rest of the environment.

5/06/2026 - Tristan

- Cleaned and organised the project assets by removing unused files, renaming texture files, and sorting materials/textures into their correct folders.

- Finalised the project hierarchy and folder structure to make the Unity scene easier to manage and reduce clutter before submission.